Smart Hospital Management System

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

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1. INTRODUCTION

1.1 Purpose

The purpose of this document is to describe the functional and non-functional requirements of the Smart Hospital Management System (SHMS). This system integrates chatbot-based symptom analysis, doctor recommendation, appointment scheduling, emergency ambulance services, and secure medical record management.

1.2 Scope

The system aims to:

- Provide symptom-based doctor recommendation
- Enable appointment booking and management
- Detect emergency cases and assign ambulances
- Support real-time ambulance tracking
- Securely manage medical records
- Implement strict role-based access control

1.3 Definitions, Acronyms, Abbreviations

Term	Meaning
SHMS	Smart Hospital Management System
RBAC	Role-Based Access Control
ICU	Intensive Care Unit
GPS	Global Positioning System
EHR	Electronic Health Record

1.4 References

- IEEE Standard for Software Requirements Specification

1.5 Overall Description

The Smart Hospital Management System is designed to improve hospital efficiency and patient care by automating doctor recommendations, managing appointments, coordinating emergency ambulance services, and securing medical records using encryption and access control mechanisms.

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2. OVERALL DESCRIPTION

2.1 Product Perspective

The system is a standalone web application consisting of:

- Frontend user interface
- Backend server logic
- MySQL database
- Secure file storage system
- GPS and Maps API integration

2.2 Product Functions

The system shall:

1. Allow user registration and login.
2. Provide role-based dashboards.
3. Accept patient symptoms through chatbot.
4. Analyze symptoms and classify urgency.
5. Recommend suitable doctors.
6. Enable appointment booking and management.
7. Trigger emergency handling for critical cases.
8. Assign nearest ambulance automatically.
9. Notify ICU doctors in emergency situations.
10. Allow doctors to upload prescriptions and reports.
11. Store medical records securely with encryption.
12. Allow admin to manage doctors.
13. Provide limited guest access.

2.3 User Characteristics

User	Characteristics
Patient	Basic computer knowledge
Doctor	Professional user with system familiarity
ICU Doctor	Handles emergency cases
Ambulance Staff	Basic system interaction skills

Admin	Technical knowledge of system management
Guest	Limited access user

2.4 User Constraints

- Users must have internet access.
- Location permission must be granted for ambulance services.
- Valid login credentials are required for full access.
- Guest users have limited functionality.

2.5 Assumptions & Dependencies

- Users provide accurate symptom information.
- GPS services are available.
- Hospital emergency infrastructure exists.
- Server and database are operational.

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3. SPECIFIC REQUIREMENTS

3.1 Interface Requirements

3.1.1 External Interface

- Web-based user interface accessible through browsers.
- Responsive design for mobile and desktop devices.

3.1.2 Hardware Interface

- Smartphone or desktop system.
- GPS-enabled device for location tracking.
- Server for hosting backend and database.

3.1.3 Software Interface

- MySQL Database
- Secure file storage system
- Maps API for location services
- Web server environment

3.1.4 Communication Interface

- HTTPS secure communication protocol
- REST APIs for data exchange
- Real-time updates for emergency notifications

3.2 Functional Requirements

3.2.1 Use Case Model / Information Flows

Actors and their interactions:

Patient:

- Register/Login
- Enter symptoms
- Book appointment
- Request ambulance
- View/download medical records
- Submit feedback

Doctor:

- Login
- View appointments
- Upload prescriptions
- Access patient records

ICU Doctor:

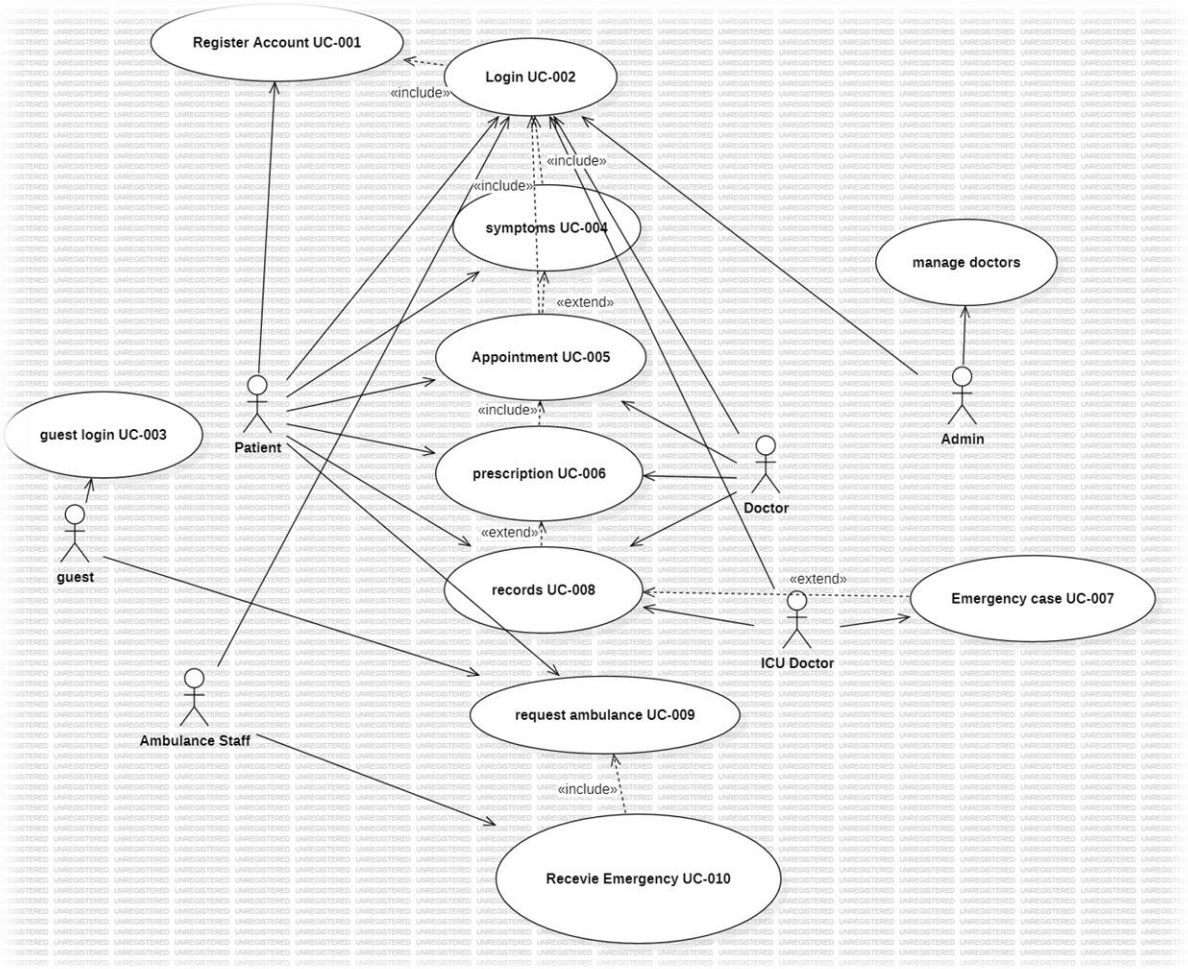
- Receive emergency alerts
- View patient emergency details

Ambulance Staff:

- Receive emergency case
- Update vitals
- Upload emergency files

Admin:

- Manage doctors
- Monitor system activity



3.2.2 Use Case Specifications / Process Description

Appointment Booking

Precondition: Patient logged in.
Process:

1. Patient enters symptoms.
2. System analyzes symptoms.
3. Recommended doctors displayed.
4. Patient selects time slot.
5. Appointment confirmed.

Postcondition: Appointment stored in database.

Emergency Case

Precondition: Symptoms classified as critical or ambulance requested.
Process:

1. Emergency alert generated.

2. Nearest ambulance assigned.
3. ICU doctor notified.
4. Ambulance staff updates patient vitals.

Postcondition: Emergency case recorded.

3.2.3 Analysis Classes / Data Dictionary

Main Entities:

- User (UserID, Name, Email, Password, Role)
- Patient (PatientID, Age, MedicalHistory)
- Doctor (DoctorID, Specialization, Availability)
- Appointment (AppointmentID, Date, Time, Status)
- Ambulance (AmbulanceID, Location, Status)
- MedicalRecord (RecordID, FilePath, EncryptedKey)

3.3 Performance Requirements

- System shall support multiple concurrent users.

3.4 Logical Database Requirements

- Database shall maintain normalized tables.
- Medical records shall be encrypted.
- Backup and recovery mechanisms shall be implemented.
- Role-based access queries shall be enforced.

3.5 Design Constraints

- Role-Based Access Control (RBAC) mandatory.
- End-to-end encryption for medical files.
- HTTPS communication required.
- Compliance with healthcare data privacy standards.

3.6 Software System Attributes

3.6.1 Reliability

3.6.2 Availability

- 24/7 system access.
- Emergency services always active.

3.6.3 Security

- Encrypted medical file storage.
- Secure authentication mechanisms.
- Restricted role-based access control.

3.6.4 Maintainability

- Modular system architecture.
- Easy debugging and updates.

3.6.5 Portability

- Compatible with major web browsers.
- Deployable on cloud or local servers.

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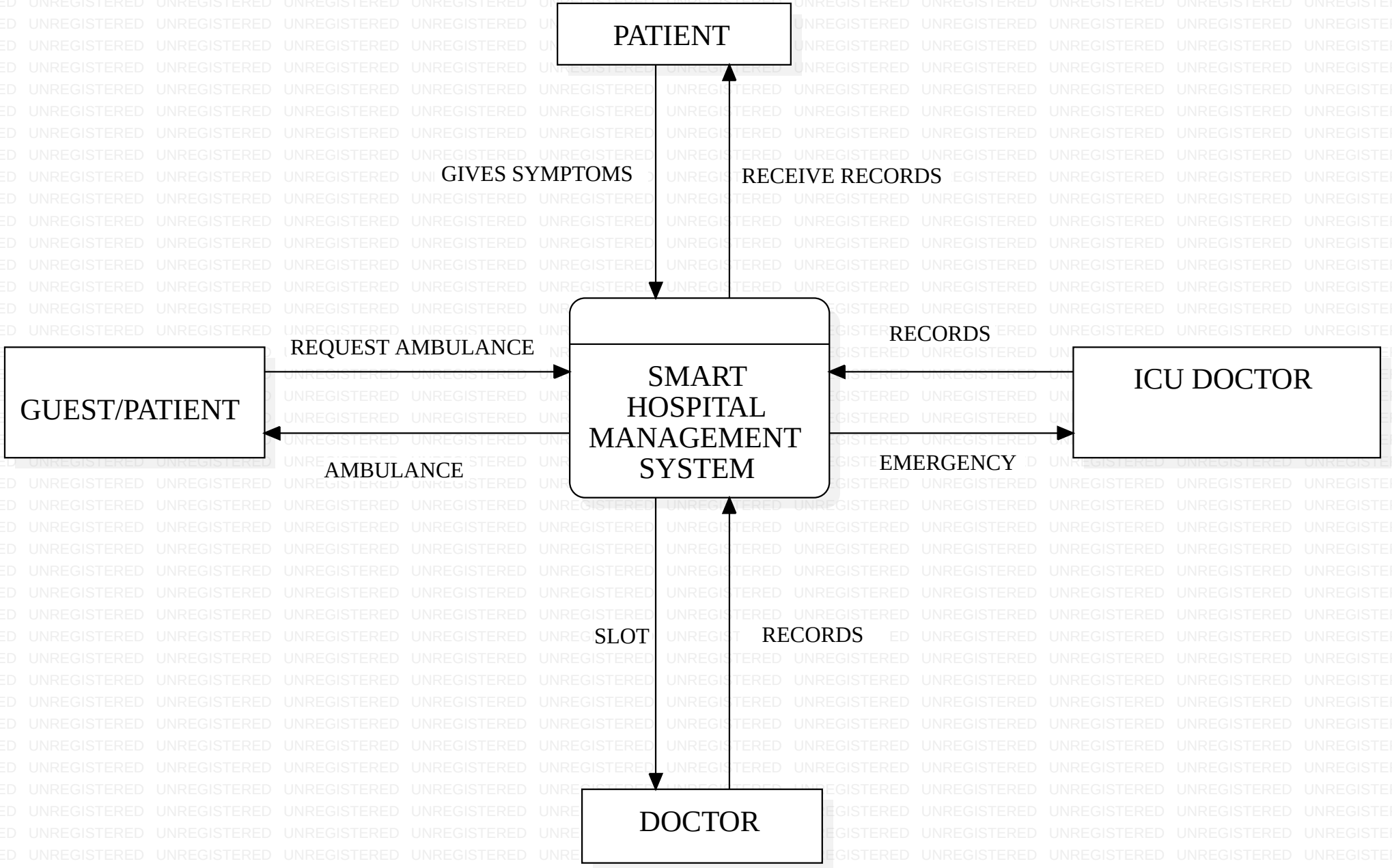
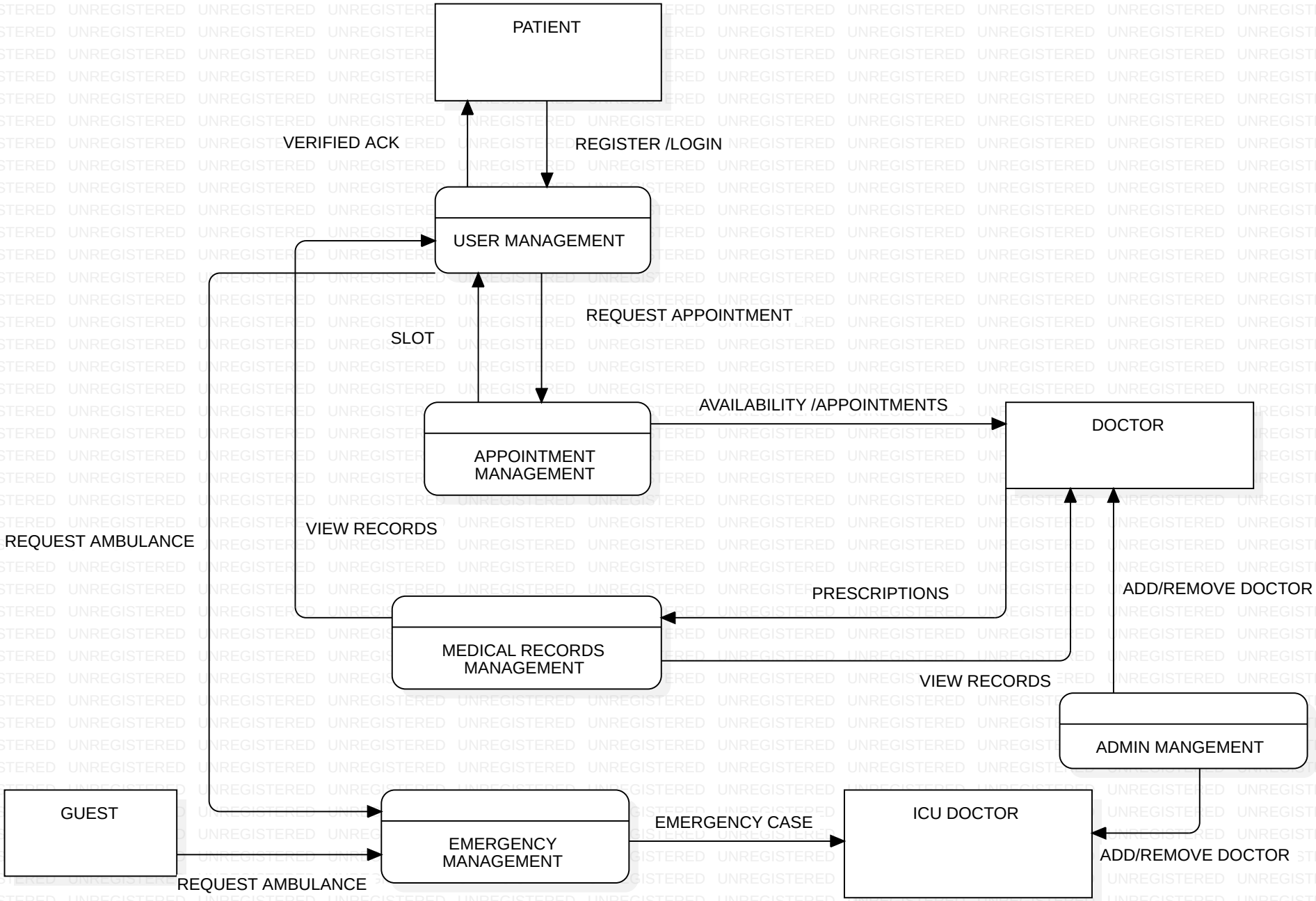
4. SUPPORTING INFORMATION

- Use Case Diagram
- DFD Level 0, Level 1, Level 2

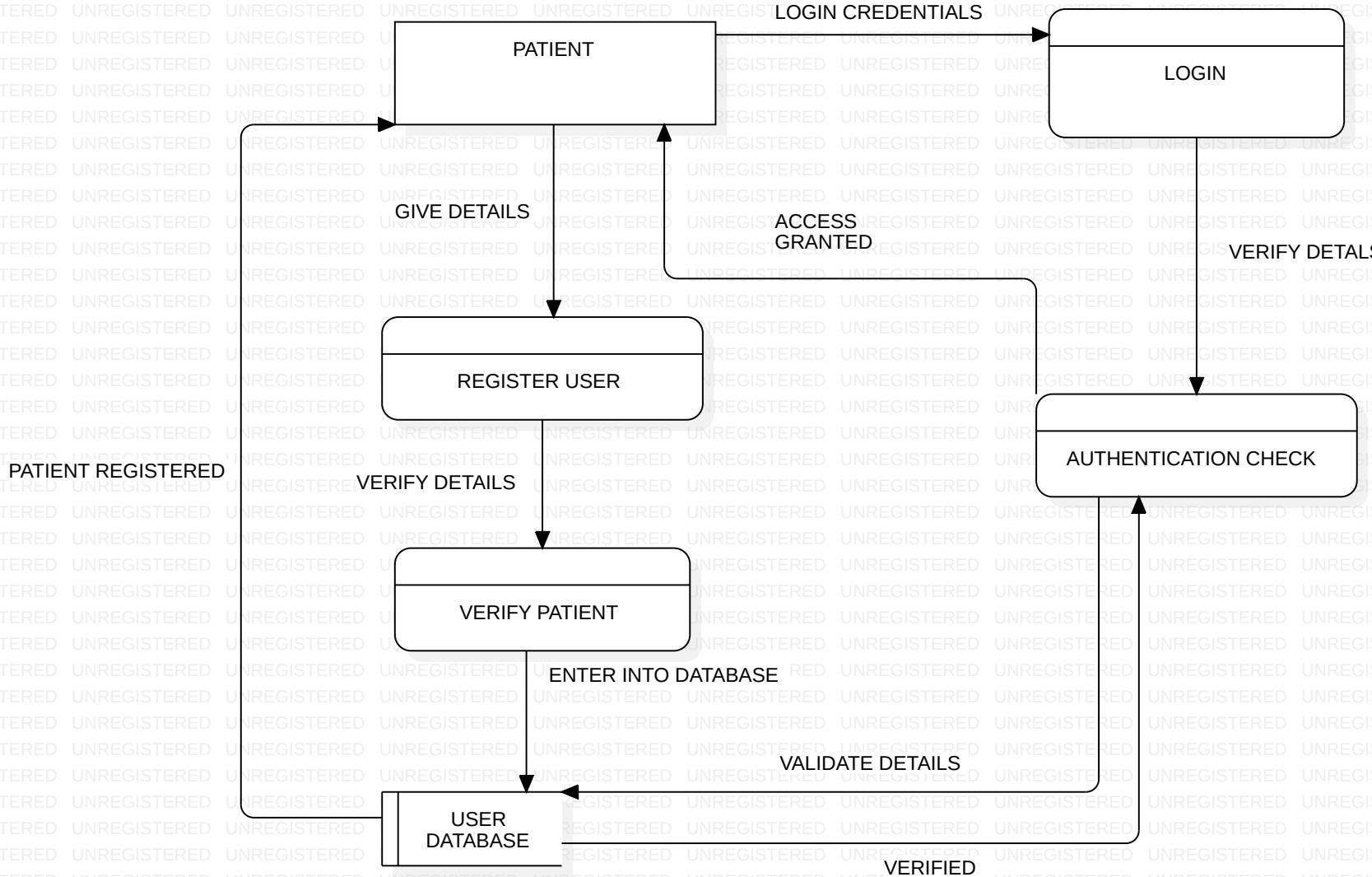
DataFlowModel1::CONTEXT LEVEL



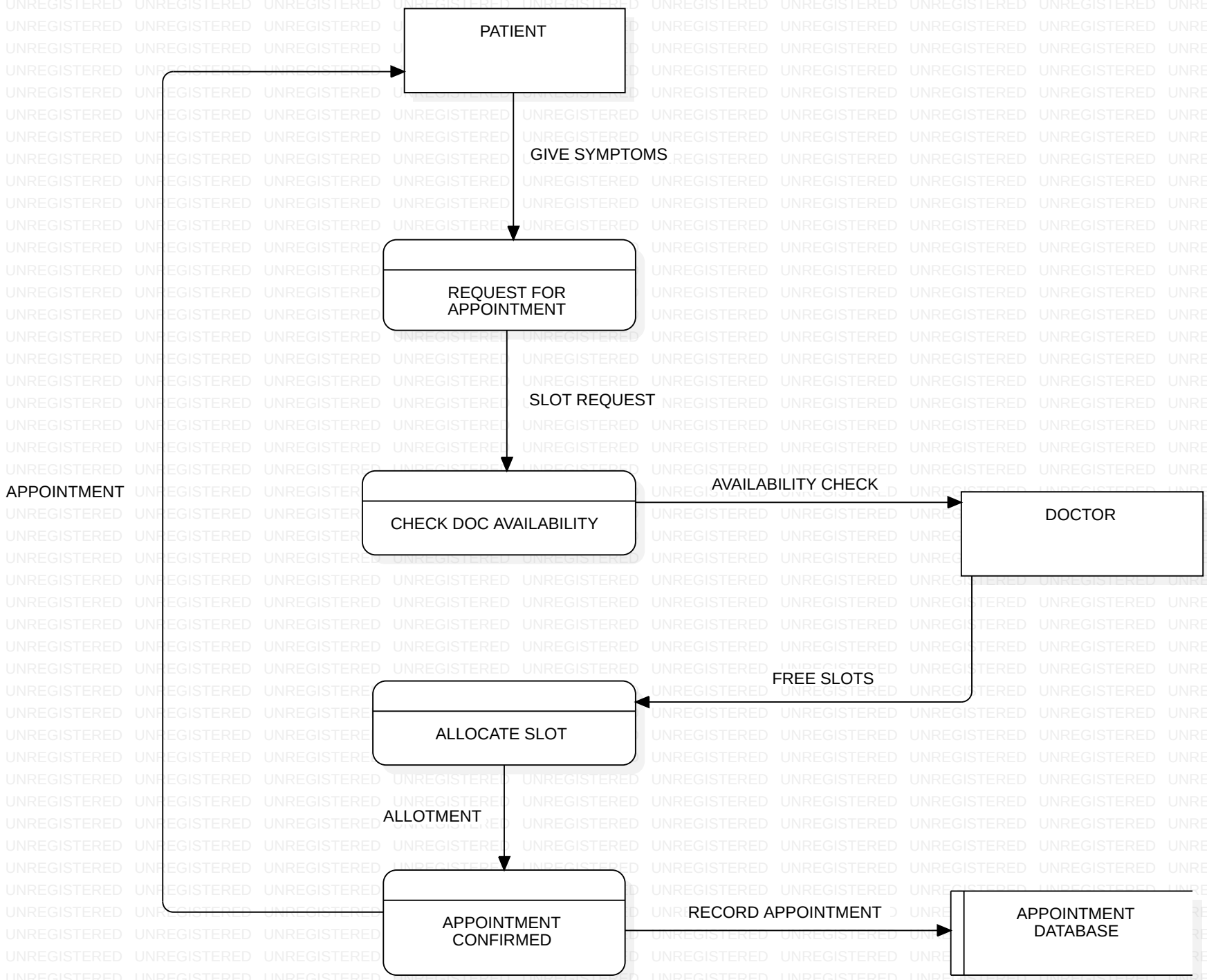
DataFlowModel2::FIRST LEVEL



DataFlowModel3::LEVEL 2 USER MANAGEMENT



DataFlowModel4::LEVEL 2 APPOINTMENT MANAGEMENT



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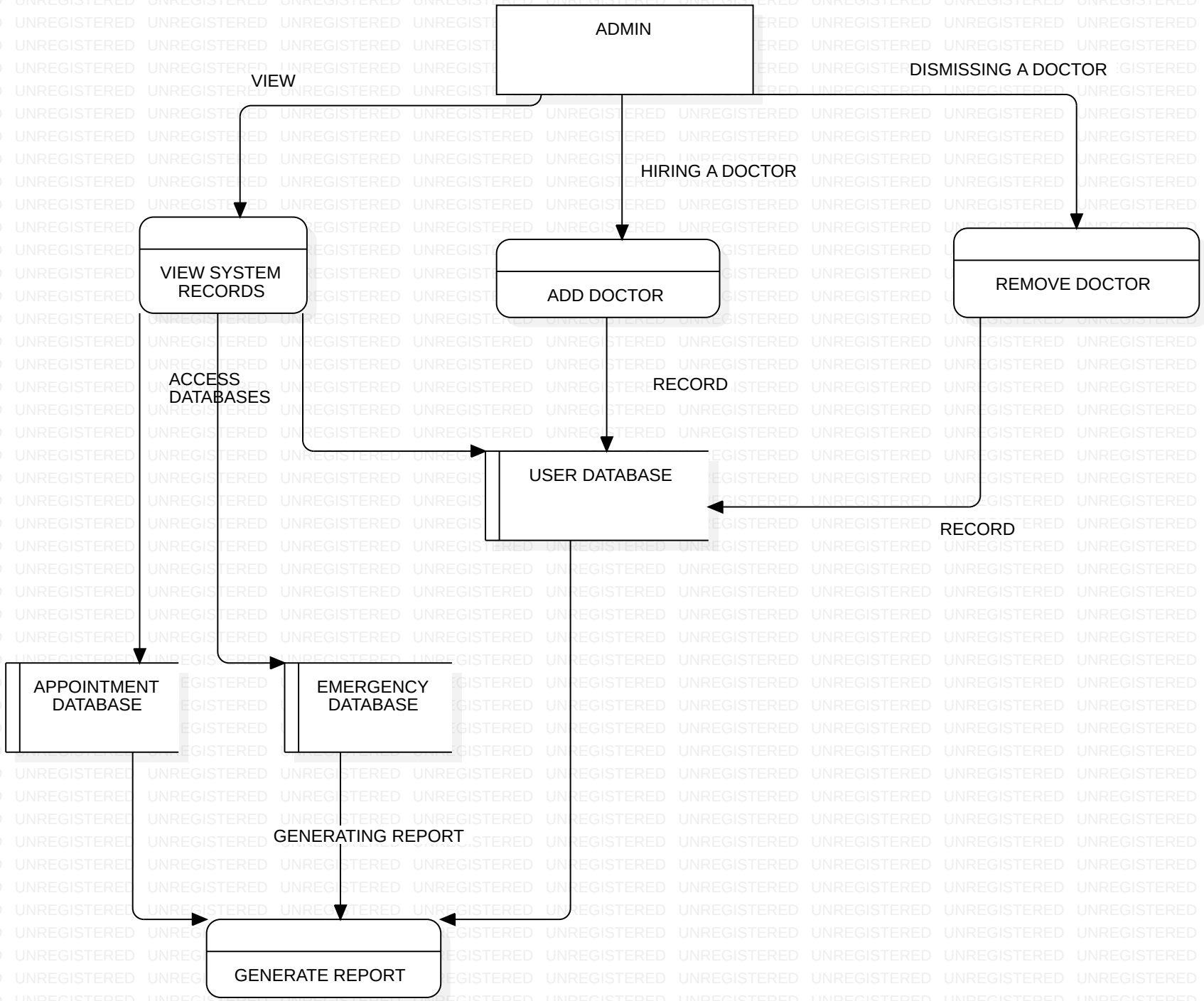
DataFlowModel5::LEVEL 2 ADMIN MANAGEMENT



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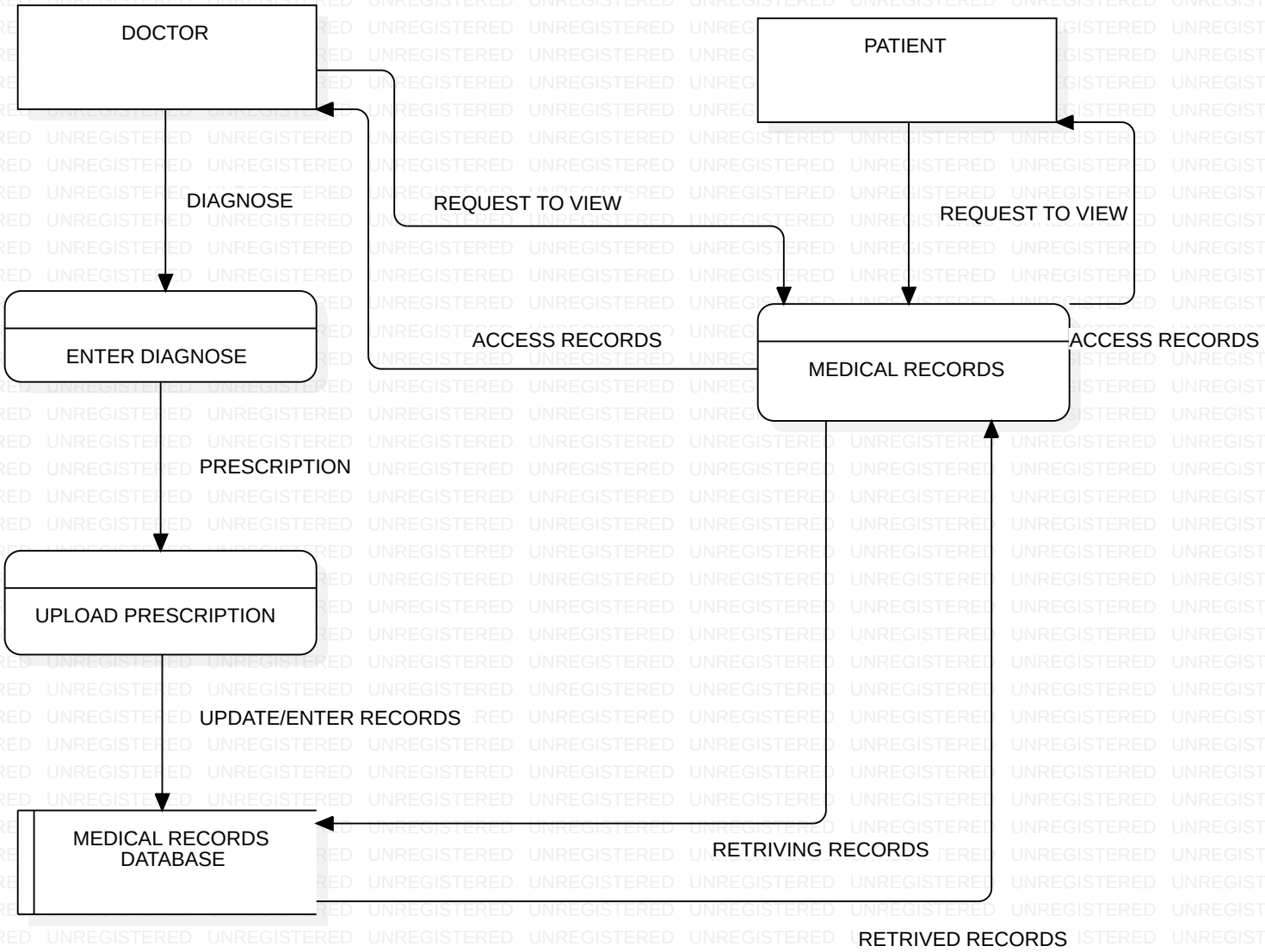
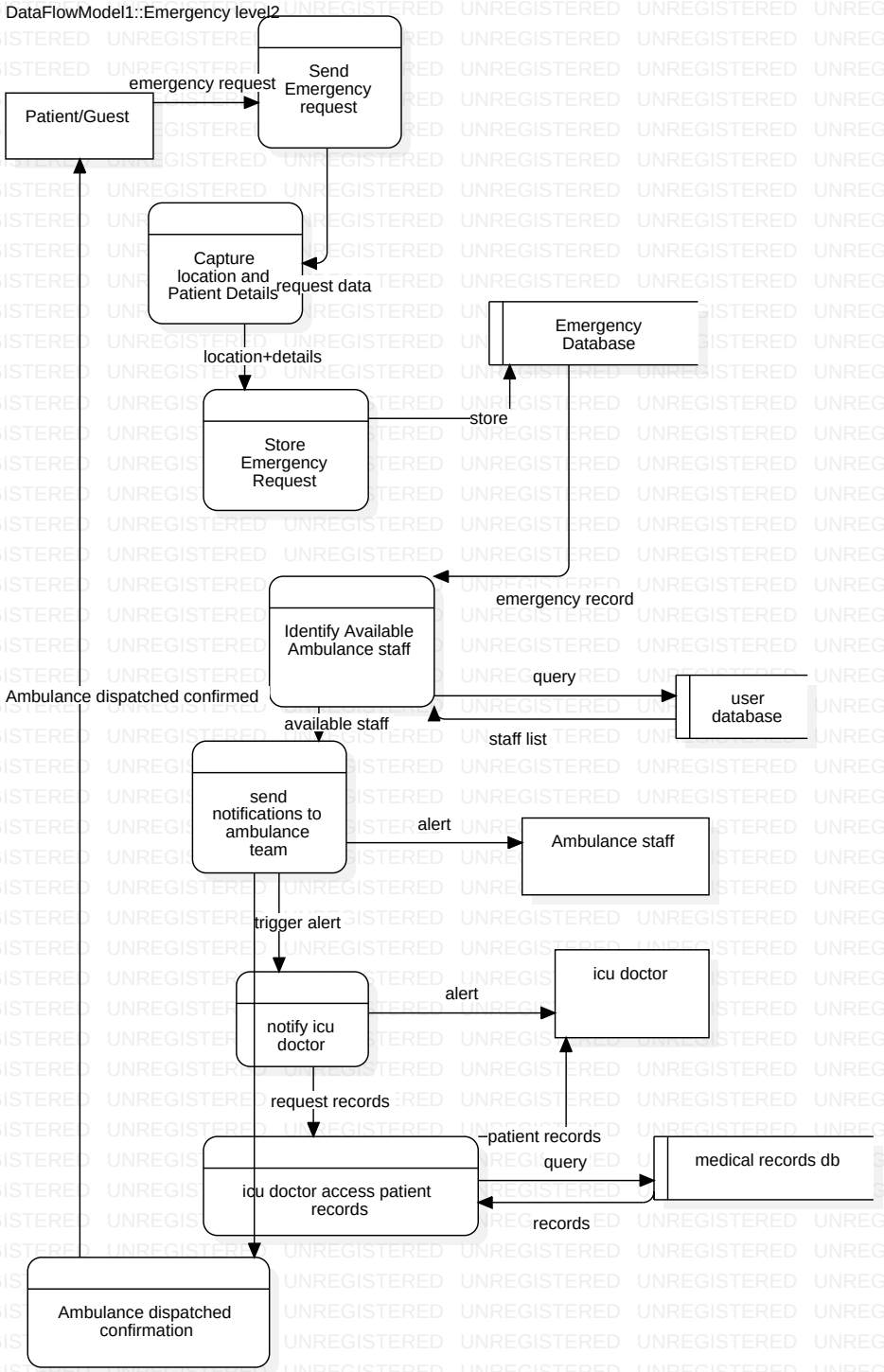
DataFlowModel6::LEVEL 2 MEDICAL RECORDS MANAGEMENT



UNREGISTERED


DataFlowModel1::Emergency level2
Patient/Guest
emergency request
Send Emergency request
Capture location and Patient Details
request data
location+details
Emergency Database
Store Emergency Request
store
emergency record
Identify Available Ambulance staff
query
user database
staff list
Ambulance dispatched confirmed
available staff
send notifications to ambulance team
alert
Ambulance staff
trigger alert
icu doctor
notify icu doctor
alert
request records
patient records
query
medical records db
icu doctor access patient records
records
Ambulance dispatched confirmation